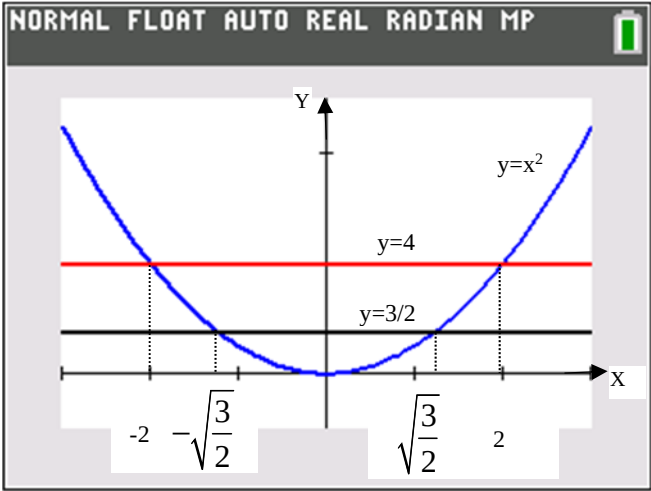
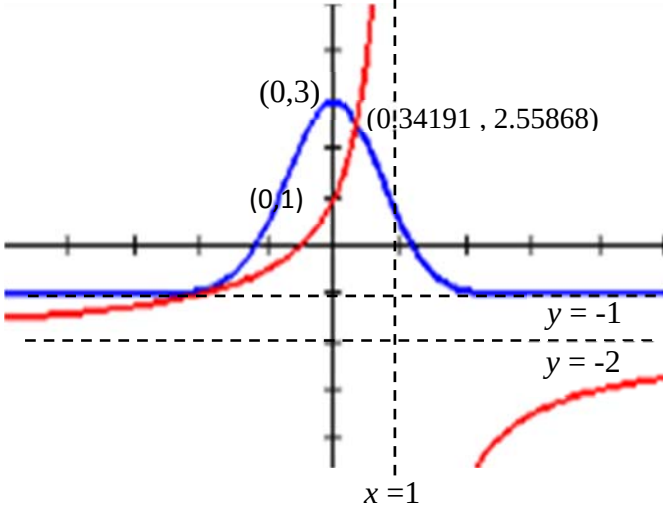


Qtn	Solution
1(a) (i)	Let t, p and c be the weights (in kg) of tomatoes, pumpkins and cherries harvested respectively. $t + p + c = 23 \text{ ----- (1)}$ $3t + 5p + 16c = 154 \text{ ----- (2)}$ $15t + kp + 45c = 545 \text{ ----- (3)}$ $35t + 50p + 100c = 735 + 545 = 1280 \text{ ----- (4)}$ Solving (1), (2) and (4) using GC, $t = 8, p = 10$ and $c = 5$ Substituting into (3), $15(8) + 10k + 45(5) = 545 \Rightarrow k = 20$
2(i)	<u>Method 1</u> $y = x^2 - 2x + 2 \Rightarrow y = (x-1)^2 + 1$ Since $(x-1)^2 \geq 0$ for all values of x, $\therefore y = x^2 - 2x + 2$ is always positive for all real values of x. <u>Method 2</u> For $y = x^2 - 2x + 2$, $D = (-2)^2 - 4(1)(2) = -4 < 0$ Since coefficient of $x^2 > 0$, $y = x^2 - 2x + 2$ is always positive for all real values of x. $\frac{x}{x-4} - \frac{1}{2x-3} = \frac{x(2x-3) - (x-4)}{(x-4)(2x-3)}$ $= \frac{2(x^2 - 2x + 2)}{(x-4)(2x-3)} \geq 0$ Since $x^2 - 2x + 2$ is always positive, we thus solve $(x-4)(2x-3) > 0$ $\Rightarrow x < \frac{3}{2} \text{ or } x > 4$

2(ii)	From part (i) and replacing x with x^2, we have if $\frac{x^2}{x^2-4} - \frac{1}{2x^2-3} < 0$, then $\frac{3}{2} < x^2 < 4 \quad \dots \dots (1)$ Solving (1), we have:  $x^2 = \frac{3}{2} \Rightarrow x = \pm\sqrt{\frac{3}{2}}$ $x^2 = 4 \Rightarrow x = \pm 2$ Thus, the solution is $-2 < x < -\sqrt{\frac{3}{2}} \text{ or } \sqrt{\frac{3}{2}} < x < 2$ or $-2 < x < -1.22 \text{ or } 1.22 < x < 2$
3(a)	<u>Method 1</u> $\frac{d}{dx}(\ln\sqrt{2x})^3 = \frac{d}{dx}\left(\frac{1}{2}\ln(2x)\right)^3$ $= \frac{1}{8}\left(\frac{d}{dx}(\ln(2x))^3\right)$ $= \frac{3}{8}\left(\frac{2}{2x}\right)(\ln(2x))^2$ $= \frac{3}{8x}(\ln(2x))^2$ <u>Method 2</u>

	$\frac{d}{dx}(\ln\sqrt{2x})^3$ $= 3(\ln\sqrt{2x})^2 \left(\frac{1}{\sqrt{2x}} \right) \left(\frac{1}{2}(2x)^{-\frac{1}{2}} \times 2 \right)$ $= \frac{3}{2x}(\ln\sqrt{2x})^2$
3b(i)	$y = \frac{a}{2x-a}$ $\frac{dy}{dx} = -\frac{2a}{(2x-a)^2}$ At point A, $\frac{dy}{dx} = -\frac{2}{a}$ Equation of tangent at A is $y-1 = -\frac{2}{a}(x-a)$ $y = -\frac{2}{a}x + 3$
b(ii)	When $x = 0, y = 3$. Coord of Q is (0, 3) When $y = 0, x = \frac{3}{2}a$. Coord of P is $\left(\frac{3a}{2}, 0\right)$ Length of PQ = 7 $3^2 + \left(\frac{3a}{2}\right)^2 = 7^2$ $a = \pm \frac{4}{3}\sqrt{10}$ Since $a > 0, \therefore a = \frac{4}{3}\sqrt{10}$
4(a)	$\int_1^2 \frac{1}{2x-1} dx = \int_1^5 \frac{k}{\sqrt{2x-1}} dx$ $\frac{1}{2} [\ln 2x-1]_1^2 = k [\sqrt{2x-1}]_1^5$ $\frac{1}{2} \ln 3 = 2k$ $k = \frac{1}{4} \ln 3$

4(b) (i)	
b(ii)	x – coordinate of pt of intersection is 0.34191 (5 d.p.)
b(iii)	Area required = $\int_0^{0.34191} \left(4e^{-x^2} - 1 - \frac{2x+1}{1-x} \right) dx$ $= 0.4028$ (4 d.p.)
5(a)	$\begin{aligned} & \frac{d}{dx} \left(x\sqrt{r^2 - x^2} \right) \\ &= \frac{d}{dx} \left(\sqrt{r^2 x^2 - x^4} \right) = \frac{1}{2} (r^2 x^2 - x^4)^{-\frac{1}{2}} (2r^2 x - 4x^3) \\ &= \frac{r^2 x - 2x^3}{\sqrt{r^2 x^2 - x^4}} = \frac{x(r^2 - 2x^2)}{x\sqrt{r^2 - x^2}} \\ &= \frac{r^2 - 2x^2}{\sqrt{r^2 - x^2}} \end{aligned}$
5(b) (i)	By Pythagoras' $AB = x - \sqrt{r^2 - x^2}$ $\begin{aligned} A &= 2x \times 2\sqrt{r^2 - x^2} + (2x - 2\sqrt{r^2 - x^2}) \times 2\sqrt{r^2 - x^2} - (1) \\ &= 4x\sqrt{r^2 - x^2} + 4x\sqrt{r^2 - x^2} - 4(r^2 - x^2) \\ &= 8x\sqrt{r^2 - x^2} - 4r^2 + 4x^2 \end{aligned}$
(ii)	$\begin{aligned} A &= 8x\sqrt{r^2 - x^2} - 4r^2 + 4x^2 \\ \frac{dA}{dx} &= 8 \times \frac{r^2 - 2x^2}{\sqrt{r^2 - x^2}} + 8x \\ \text{When } \frac{dA}{dx} &= 0, \end{aligned}$

	$r^2 - 2x^2 + x\sqrt{r^2 - x^2} = 0$ $x\sqrt{r^2 - x^2} = 2x^2 - r^2 \text{ ---- (2)}$ $x^2 (r^2 - x^2) = 4x^4 - 4r^2x^2 + r^4$ $5x^4 - 5r^2x^2 + r^4 = 0$												
(iii)	$5x^4 - 125x^2 + 625 = 0$ $x = 4.25 \text{ or } x = 2.63$ In equation (2), $2x^2 - r^2 \geq 0 \Rightarrow x^2 \geq \frac{1}{2}r^2$. That is $x^2 \geq \frac{1}{2}(5^2) = 12.5$. Hence, $x = 2.63$ is rejected. $x = 4.25$ <table border="1"><tr><td>x</td><td>4.25^-</td><td>4.25</td><td>4.25^+</td></tr><tr><td>$\frac{dA}{dx}$</td><td>+ve</td><td>0</td><td>-ve</td></tr><tr><td>Slope</td><td>/</td><td>-</td><td>\</td></tr></table> Hence, when $x = 4.25$, A is max	x	4.25^-	4.25	4.25^+	$\frac{dA}{dx}$	+ve	0	-ve	Slope	/	-	\
x	4.25^-	4.25	4.25^+										
$\frac{dA}{dx}$	+ve	0	-ve										
Slope	/	-	\										
6(i)	Required probility $= \frac{26 \times 26 \times 26 \times 26}{36 \times 36 \times 36 \times 36} = \left(\frac{26}{36}\right)^4 = 0.272$												
6(ii)	Required probability $= \frac{{}^{26}C_3 \times {}^{10}C_1 \times 4!}{36^4} = \frac{1625}{4374} \text{ or } 0.372$												
7	Let $X \text{ cm}^3$ and $Y \text{ cm}^3$ be the volumes of glass and wood in a paperweight respectively. $X \sim N(56.5, 2.9^2)$ $Y \sim N(38.4, \sigma^2).$ $\therefore X + Y \sim N(94.9, 2.9^2 + \sigma^2)$ $P(X + Y > 100) = 0.05$ $\Rightarrow P(X + Y \leq 100) = 0.95$ $\Rightarrow P\left(Z \leq \frac{100 - 94.9}{\sqrt{2.9^2 + \sigma^2}}\right) = 0.95$ $\Rightarrow \frac{5.1}{\sqrt{2.9^2 + \sigma^2}} = 1.64485$ $\Rightarrow 2.9^2 + \sigma^2 = \left(\frac{5.1}{1.64485}\right)^2$												

	$\Rightarrow \sigma = \sqrt{1.20363} = 1.09710 = 1.10 \text{ (to 3 sf)}$
8(i)	Using mean = np $12\left(\frac{p}{100}\right) = 2.4$ $\Rightarrow p = 20$
(ii) (a)	Let X be the no. of rotten mangoes out of 12 mangoes. $X : B(12, 0.2)$ $P(X \geq 2) = 1 - P(X \leq 1) = 0.725$
(b)	Let Y be the no. of rotten mangoes out of 11 mangoes $Y : B(11, 0.2)$ Required probability $= 0.2 \times P(Y = 1)$ $= 0.04724$ $= 0.0472 \text{ (to 3 s.f)}$
	Answer to part (b) is smaller than answer to (a) because event in (b) is a proper subset of event in (a). Event in (a) includes cases where 2 rotten mango can be any 2 of the 12 whereas for event (b), the second rotten mango must be the 10th mango.
(iii)	Let $\bar{X}$ be the mean no. of rotten mangoes in a box. Since n is large, by the Central Limit Theorem, $\bar{X} \sim N\left(2.4, \frac{2.4 \times 0.8}{60}\right) \text{ approximately.}$ $P(\bar{X} < 2) = 0.0127$
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(ii)	Product moment correlation coefficient $r = 0.969$ which indicates a strong positive linear correlation between the braking distance, y metres, and different speeds, $x \text{ kmh}^{-1}$, when the brakes are applied.
(iii)	From GC, regression line is $y = 3.2143x - 107.143 \Rightarrow y = 3.21x - 107$
(iv)	When $x = 100$, $y = 3.2143 \times 100 - 107.143 = 214.287 = 214$ (3sf) $x = 100$ is within the data range $[30, 130]$ and $r = 0.969$ is close to 1, suggesting strong positive linear correlation between x and y. Therefore the estimated value of $y = 214$ obtained should be reliable. When $x = 150$, $y = 3.2142 \times 150 - 107.142 = 374.988 = 375$ (3sf) $x = 150$ is outside the data range $[30, 130]$ which means the estimated braking distance of 375 m obtained from the equation of regression line may be unreliable.
(v)	$u = 80, v = 150$
10(i)	$P(C) = \frac{16 + 2x}{48 + 3x}$ $P(C H) = \frac{2x}{32 + 2x} = \frac{x}{16 + x}$
(ii)	$\frac{16 + 2x}{48 + 3x} = \frac{x}{16 + x}$ $(16 + 2x)(16 + x) = x(48 + 3x)$ $256 + 48x + 2x^2 = 48x + 3x^2$ $x^2 = 256$ $x = 16$
(iii)	$P(C' S')$ is the probability that a randomly selected student has not visited Chiang Mai given that he/she has not visited Surabaya. $P(C' S') = \frac{P(C' \cap S')}{P(S')} = \frac{20/96}{47/96}$ $= \frac{20}{47} \text{ or } 0.426$
(iv)	Since $P(C' S') = \frac{20}{47} \neq P(C') = \frac{1}{2}$ $\therefore C$ and S are not independent.
11(i)	Let X be the weight of a jar and μ denotes the population mean weight of a jar in grams $H_0 : \mu = 502$ $H_1 : \mu < 502$

(ii)	Under H_0, Test statistic $Z = \frac{\bar{X} - 502}{\sqrt{\frac{10.5^2}{50}}} : N(0,1)$ At 1% level of significance, under H_0, since $n = 50$ is large, by Central Limit Theorem $\bar{X} : N\left(502, \frac{10.5^2}{50}\right)$ approximately $p = 0.0217 > 0.01$, we do not reject H_0 and conclude that at the 1% level of significance, there is insufficient evidence to claim that the mean weight of a jar is less than 502 g.
(iii)	There is a probability of 0.01 that we wrongly conclude the mean weight of a jar is less than 502 g when in fact the mean weight of a jar is 502 g.
(iv)	Unbiased estimate for the population mean $= \frac{\sum (x - 502)}{60} + 502 = 498.3$ Unbiased estimate for the population variance $= \frac{1}{59} \left(\sum (x - 502)^2 - \frac{(\sum (x - 502))^2}{60} \right)$ $= 128.47$
(v)	$H_0 : \mu = 502$ $H_1 : \mu \neq 502$ Under H_0, since $n = 60$ is large, by Central Limit Theorem $\bar{X} : N\left(502, \frac{128.47}{60}\right)$ approximately Under H_0, Test statistic $Z = \frac{\bar{X} - 502}{\sqrt{\frac{128.47}{60}}} : N(0,1)$ approximately By GC, $p\text{-value} = 0.011453$ Since H_0 is rejected, $\frac{\alpha}{100} \geq 0.011453$ Set of values = $\{\alpha \in \mathbb{R} : 1.15 \leq \alpha \leq 100\}$
12	Let X and Y be the weight (in kg) of a packet of type A and type B strawberries respectively. $X : N(1.2, 0.2^2)$ $Y : N(1.1, 0.1^2)$ (i)

	$P(X \geq 1.2)P(Y \geq 1.2) = 0.0793$ (ii) Let $T = X + Y : N(1.2 + 1.1, 0.2^2 + 0.1^2)$ $T : N(2.3, 0.05)$ $P(T < 2.2) = 0.32736$ $P(2.2 < T < m) = 0.45$ $P(T < m) - P(T < 2.2) = 0.45$ $P(T < m) = 0.32736 + 0.45 = 0.77736$ $\therefore m = 2.47$ (iii) Let A and B be the cost of one packet of type A and type B strawberries respectively. $A = 20X : N(20 \times 1.2, 20^2 \times 0.2^2) = N(24, 16)$ $B = 26Y : N(26 \times 1.1, 26^2 \times 0.1^2) = N(28.6, 6.76)$ Let C be the total cost for 2 packets of strawberries picked by Audrey. $C = A + B : N(24 + 28.6, 16 + 6.76) = N(52.6, 22.76)$ $P(C \leq 50) = 0.29288 = 0.293(\text{to } 3 \text{ s.f})$ (iv) Let M be the average weight of the three packets of type B strawberries bought by Jackson. $M = \frac{Y_1 + Y_2 + Y_3}{3}$ $M : N\left(\frac{3(1.1)}{3}, \frac{3(0.1^2)}{3^2}\right) = N\left(1.1, \frac{0.1^2}{3}\right)$ $X - M : N\left(1.2 - 1.1, 0.2^2 + \frac{0.1^2}{3}\right) = N\left(0.1, \frac{0.13}{3}\right)$ $P(0 \leq X - M < 0.2) = 0.36905 = 0.369(\text{to } 3 \text{ s.f})$
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